# Using No-code Platform to Accelerate Digital Transformation – The Study for Feasibility Analysis and Solution Development of Performance Visualization Cloud-computing Management System (PVCMS)

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## ABSTRACT

This study is based on government's digital transformation policy and the industry's digital development trend. Micro-Small Enterprises(MSEs) are the target customers to meet the organizational productivity management needs of key performance indicators (KPI) and project management (PjM) as the research subject, combined with management by objectives (MBO) and performance appraisal organizational management practice as a methodology, using Google's No-code smart cloud-computing service as a practical platform, integrating data input/processing/analysis/ visualization (D/IPAV®) workflows to customized develop the "Performance Visualization Cloud-computing Management System" (PVCMS) of the organization's operational data monitoring and decision-making digital dashboard. This project sets PVCMS as the core product and develops a set of indicators construction, analysis and visual monitoring that meet the performance management needs of the industry. Diagnosis and Consultant model as a solution, to follow-up actual commercialization standard operating procedure to be followed, and quick import templates for various performance management visualization modules. Its operation process is mainly divided into five stages: (1). Define performance management requirements, (2). Establish KPIs, (3). Link Indicator with performance evaluation, (4). Build performance management system, (5). Track Indicators and decision-making application.

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## CCS CONCEPTS

• **General and reference**; • **Cross-computing tools and techniques**; • **Measurement**; • **Information systems**; • **Information systems applications**; • **Computing platforms**; • **Applied computing**; • **Document management and text processing**; • **Document capture**; • **Document analysis**;

## KEYWORDS

Key Performance Indicators (KPI), No-code Platform, Digital Transformation, Performance Visualization Cloud-computing Management System (PVCMS), Cloud-computing service

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## 1 INTRODUCTION

In this section, the researcher explains the background, objectives, and benefits of this research.

### 1.1 Trend for Digital Transformation

Gartner, an international research and consulting agency, recently proposed ten strategic technology trends that enterprises must understand in 2020, namely hyper-automation, multiple experiences, democratization of expertise and technology, enhanced human empowerment, transparency and traceability, and more. Powerful edge computing, decentralized cloud, automated objects, practical blockchain, and AI security. Among them, for the democratization of professional knowledge and technology, Gartner predicts that by 2023, this trend will accelerate development in four major areas: application development, data and analysis technology, design, and knowledge [28]

Under the influence of the COVID-19 epidemic, many teams are racing to reshape the enterprise organization through "digital transformation". In particular, they continue to focus on the complex development of large-scale systems, but find it difficult to make rapid progress. As experienced software engineers give priority to develop complex core products, so small software development projects and requirements quickly accumulate. In contrast, the introduction of the No-Code development platform allows the business requirements team to develop the required apps without the need for programming, which greatly improves the development speed [18].

The researcher pointed out that digital transformation is not a new strategy. With the evolution of industrial technology and internal and external business conditions, digital transformation in different eras has different business strategy significance [21]. The early digital transformation is mainly to improve the internal operation efficiency of the company through the introduction of computer information systems, and then combine the company's business process management (BPM) and the indispensable MRP, ERP and MES systems to bring dividends to the industry in digital transformation. The recent new wave of digital transformation is the application of cloud technology, big data analysis, IoT, artificial intelligence and other technologies to assist industrial upgrading and transformation. With the increasing number of successful cases of digital transformation and the recent upsurge in the application of big data and artificial intelligence, everyone has higher expectations for digital transformation.

The development of business Intelligence field can't ignore the cloud computing trends. There are many benefits from using the cloud computing for business intelligence. It influences the way business intelligence software projects are managed which it provides a virtually unlimited pool of computing power, storage space and memory for the business intelligence infrastructure [33]. Hybrid cloud is a type of cloud computing infrastructure that is composed of both public and private cloud. When resources in private cloud are not sufficient, resources from public cloud are taken on lease, to complete the execution of an application. Hence in hybrid cloud, the benefits of both public cloud and private cloud can be utilized by an application. Hybrid cloud has the advantages of optimal resource provisioning, scalability and pay-as-you-use benefits [24].

Cloud computing is an upcoming trend in the field of distributed computing in recent years. In cloud, resources are provided as a service in the form of virtual machines to its clients based on demand, which may result in overutilization or underutilization of servers. Cloud must accommodate changing demands for different types of processing with heterogeneous workloads and time constraints [36]. Cloud computing provides solutions as a service to meet customers' needs such as massive data storage and a lot of computing resources to process customer data efficiently [5]. Big data and Cloud computing are emerging as new promising technologies, gaining noticeable momentum in nowadays IT. Nowadays, and unprecedentedly, the amount of produced data exceeds all what has been generated since the dawn of computing; a fact which is mainly due to the pervasiveness of IT usage and to the ubiquity of Internet access [1].

While we hear stories in many news outlets of how companies are using data to succeed, academic researchers conduct empirical research and find that data-driven decisions can positively impact operational performance. This is in part due to the fact that companies are able to utilize data collected from management information systems, such as operational data from enterprise resource planning (ERP), customer relationship management (CRM) and supply chain management (SCM) systems. Enterprises can further enhance their analytical capabilities by using business intelligence (BI) systems, and use data analysis as the basis for decision-making, and have more opportunities to improve performance. More importantly, various data analysis techniques can use data from multiple sources and derive value-creating insights from these data analysis [9] [10].

### 1.2 Importance for Digitalization of Performance Management

Organizational performance management has always been an important topic of concern to business leaders and managers. Before implementing effective performance management, it is necessary to find out the associated data of the corresponding performance indicators to track. , observation, as a reference for performance management and decision-making in resource allocation and strategy development; and in a modern organization where management by objectives (MBO) is responsible, managers at all levels will also pay close attention to the performance of the department, together with individual employees performance is also related to the promotion of employees' future functions, performance bonuses and career development, promotion and rotation, etc. Key performance indicators (KPI) are all catchy, but according to the actual organization of the research team in the enterprise. Coaching experience shows that corporate KPIs and employee performance appraisal project formulation often do not have appropriate correlation, but only a form of echoing what others say.

With the changing dynamics of business models, macroeconomic trends, and digital disruption, performance management has become a hot topic of discussion in the HR function. While some of the organizations have re-modeled their overall approach to measure performance, some are currently thinking of following the industry trend and there are few that might continue with their current approach and may introduce small changes in the times to come [34]. Considering the timeliness and interactivity of performance indicators update, and when the demand for digital performance management information system is derived, under the guidance of the traditional management information system development process, it is necessary to build a large-scale information department with the ability to develop its own system. For enterprise organizations, more human and financial resources are required for technical investment and time consumption; it may also be that due to the heavy workload of the information department, the needs of the business management unit have not been placed on the development schedule, or the information department has There are doubts about many functions, and even poor communication between the two parties may delay the implementation of the management system that the property management unit desperately needs.

Performance is very important for a company or organization to achieve the goals of the company, so that employee performance needs to be improved both in the field of expertise and in the field of mastery of technology to make work easier or increase productivity in the organization or company. So far, employees are often given training to improve performance without being matched by the use of digitalization technology. Performance management is the process of consolidating goal setting, appraisal, and performance development into a single, shared system, which aims to ensure employee performance supports the company's strategic goals [39]. Meanwhile, when powered by advanced ICT, the performance measurement system matures. The design and development of the ICT platform also reinforce the organization's existing performance management practices. Empowerment is strengthened when automated collection, analysis and reporting of performance data free up middle managers' time so that they, together with operators, can drive continuous improvement.[22]

Based on the formulation of problems, objectives, theoretical foundations, hypotheses and test results conducted, it can be concluded that performance management has a positive effect on the concept of digitalization and employee performance. The concept of digitalization also positively affects employee performance [2]. As the Director and Group CHRO of Welspun Group – The researcher point that performance management systems erase some of the doubts form the past; it ensures precise performance appraisal data that is extremely critical during performance appraisals [34]. Access to their goals and evaluations for the coming year, help systems gather more feedback from co-workers and help with a positive outlook towards performance reviews. Other than accurate data, digitization helps management processes and strategic development. Tools that are driven by technology ease a manager's evaluation process and help employees become active participants in their review sessions.

About the effect of digitalization on employee performance, research results showed that digitalization affects employee performance [2]. Digitalization is related to companies that want to improve employee performance and maintain the sustainability of their organization must be able to make good decisions for the organization in it effectively reflected in the purpose or vision of the mission. The employee performance management system is a narrowed form of the broader concept of performance management as it solely applies to employees. The employee performance management system implies an HRM system that serves to continuously monitor the performance of employees. As such, it ensures that employees' efforts are aligned with the overall organizational objectives [3].

However, if it is a small and micro enterprise with streamlined personnel and small business scale, although the business owner attaches the same importance to performance management and also wants to have a performance management system, due to the realistic constraints of human and financial resources, such as senior managers The digital transformation knowledge of management cadres is generally lacking, the professional information talents are insufficient, and the information budget is low, which makes it difficult to simultaneously introduce digital performance management tools when the industrial technology changes, and it still remains in the process of error-prone, waste of manpower and difficult to reflect performance in a timely manner on manual work.

### 1.3 Objectives of this Study

Based on the trend of digital transformation, this study targets small and micro enterprises with the largest number of domestic entrepreneurs but relatively lack of resources. Main objectives of this research are as follows:

(1) To explore on the performance appraisal management system for Micro-Small Enterprises.
(2) To investigate the feasibility on cloud-computing no-code platform for performance appraisal with data input/processing/analysis/visualization process.
(3) To build a model of PVCMS focus on dashboard of KPIs combined cloud-computing no-code data platform with consultant diagnosis workflow as a KPIs management solution.

## 2 LITERATURE REVIEW

As far as this study intends to develop a solution for the diagnosis and coaching mode of performance management indicator establishment, analysis and visual monitoring, it should focus on the literature discussion in the two fields of performance management system and KPI, data visualization and dashboard. As well as studying the relevant application examples and research experience of the domestic industry, the comments are as follows:

### 2.1 Performance Management System & KPI

The goal setting method of performance evaluation is more commonly known as Management by Objectives (MBO), also known as: results management, performance management, and work planning and inspection plans. The management by objectives process generally consists of the following steps: (1). Establish a clear and precisely defined goal statement for the work an employee is to do; (2). Develop an action plan that indicates how these goals will be achieved; (3). Allow Employees implement the action plan; (4). Measure achievement of goals; (5). Take corrective action if necessary; (6). Establish new goals for the future. For an objective management system to be successful, it must meet several requirements: objectives should be measurable and measurable, and objectives that cannot be measured or confirmed should be avoided as much as possible. Objectives should also be attainable and they should be expressed in written language that is clear, precise, and unambiguous [1].

[3] pointed out in the article "Developing a KPI-driven Data Strategy: Business Strategy, Data, and Management Practices" that KPI-driven data analysis strategy development and practice can help managers find Identify priorities and priorities; and propose four steps to use data to build the right KPIs to guide better allocation of resources to meaningful measurements:(1). Start with the Industry: Designing and perfecting KPIs should start at the industry/industry level, because the similarities of transaction types, products and service offerings in the same industry can be compared for performance measurement. In addition, common data sets are often compiled in the industry to provide business leaders with meaningful benchmarks to evaluate performance and formulate operational strategies. (2). Consider the Inputs: KPIs may be

financial (such as revenue, profit, cash flow) or operations (such as supply chain, human resources, manufacturing) and related business processes (such as supplier relations, Customer relationships, production and operations, and sales and marketing) will affect the focus on certain technical systems and data in multiple ways, and consider input indicators as the starting point of the KPI process. (3). Identify and Capture the Right Data: With the proliferation of data assets in enterprise organizations, decision leaders must work closely with data analysts in the process of strategy development and implementation. Customer, product, supply chain, operation and other aspects of data, but there will be difficulties in the integration of different systems when connecting, how to connect data through API, and the future expansion of functions in process automation and real-time update, so that key users can Use this information smoothly on different vehicles or platforms. (4). Establish ownership for the KPI: When working with customers, there is a common saying "The business doesn't run inside a spreadsheet." (It can be interpreted as: Laws alone are not enough). Therefore, it is necessary to set various KPIs to the responsible department or individual employee, and it is necessary to effectively communicate and adjust management incentives, so that the corresponding departments or individuals are willing to support, so that the KPI can achieve the effect of performance management.

Olsen et al. [32] used Lohman et al. [27] to define the performance measurement system (PMS) as: Contains a set of variables (or indicators) used to quantify the efficiency and effectiveness of actions, as well as software and hardware techniques and procedures associated with data collection; and Neely et al. [30] defining a performance measurement management system (PMMS) is to encompass the interaction between the PMS and its management actions and the environment. At the same time, Olsen et al. continued the arguments of Lohman, et al. [27] and Schneiderman [35], viewing the challenge of design as an evolutionary improvement process, and proposed a method based on three criteria: causality, persistence continuous improvement and process control A framework for assessing the effectiveness of a PMS, using case studies to illustrate the application of the method and the interpretation of the results. The framework provides a simple method that organizations can easily adopt to analyze individual and group performance metrics and relate them to corporate strategic performance metrics, i.e. in line with emerging research directions, the design of the integrated PMS as a continuous improvement process.

Okwir et al. [31] argue that complexity can negatively impact the process of continuous improvement of a PMS, and previous PMS literature has argued that complexity is a result of the external environment rather than the user's reaction to that environment. Okwir et al. argue that organizations often also face internal complexities when adopting a PMS, and that, depending on the complexity of the organization's relevant members and their interactions, the introduction of a PMS into an organization may have different effects on the organization. Its research summarizes six sources of complexity: three types of roles, task, and procedural complexity in the social dimension, and three types of methodological, analytical, and technical complexity related to the technical dimension; the findings also discuss that PMS typically prolongs lag and inability to respond and remain resilient in emerging contexts. While it is useful to understand and explore all of the organizational controls that regulate PMS, organizations should still build the necessary capabilities to adapt PMS to their context.

## 2.2 Data Visualization & Dashboard

In an experimental study of U.S. Public Health Administrators, graphical data visualization was used to present performance data, especially ratio indicators compared to benchmarking reporting, compared to In terms of simply presenting performance data itself, it can facilitate managers to actually use these visual performance information to integrate into their decision-making considerations [6].

Regarding the use of data presentation methods to change personal behavior, in a basic understanding, quantitative data presentation methods can profoundly affect the reception and processing of information [6, 19]. The regular collection, reporting and discussion of goal-oriented data is a fundamental concept of performance management [7, 29]. Ideally, data will be immediate, relevant, and results-oriented, all three characteristics that help increase the relevance and validity of discussions around operational goals [8].

Lin et al. [25] used the advantages of information to obtain more time with less effort through an accurate, fast, stable and consistent monitoring system, so that hospital infection control can be more effective in responding to the epidemic and improving quality, and indirectly improve the quality of medical care and the safety of patients seeking medical treatment.

Data analysis is a science that combines data processing and the use of auxiliary tools to analyze strategies to generate decision-making plans, and then visualize the data to present the results in charts, so as to easily analyze the information contained in the report, and how to derive the needs of decision makers according to needs. The orientation of data analysis system is the value of data analysis system [19]. Data visualization aims to transform large amounts of data into simple information using statistical and graphical techniques. Since humans have the ability to quickly understand and recognize graphics, today's visualization tools convert a large amount of data into graphics through special algorithms, and use interactive technology to allow users to quickly understand the hidden knowledge in the graphics [13].

If you can make good use of analysis and visualization tools to deeply extract useful information from massive data and form conclusions, and speed up its decision-making efficiency, it will help organizational units in business management or marketing promotion. For libraries that collect various data resources, making good use of data visualization tools to mine and display the results should help library units or practitioners to think about new forms or services for adding value to data. This article quotes British data journalist David McCandless as saying: In the current environment full of data, data visualization can help people to explore data, especially when people lose their way in the sea of data, visual guidance will play a powerful role [15].

Artificial intelligence (AI) should be a term that is often mentioned in the commercial field after Big data, AIoT, FinTech, online and offline, and new retail, and is the most eagerly anticipated import project for companies. However, it often stays in the wait-and-see stage. Enterprises believe that data analysis can bring benefits, but first wait for the industry to have concrete results! Or if there are successful cases, you can refer to it and then think about whether to import it. However, in an increasingly competitive environment, consumers were originally oriented towards convenient shopping, cost-effective, and fast delivery. Coupled with the impact of COVID-19, many consumption patterns have moved towards zero-contact, take-out, online shopping, hourly delivery, etc., forcing retail Service providers are changing their familiar business models. It also prompts the industry to think about how to think about shopping needs from the perspective of consumers, predict consumption needs in advance, prepare goods early, and arrive quickly, so as to achieve a close consumption environment where people, goods, and markets are deployed ahead of time. Especially in the face of the characteristics of large, fast and complex data sources, how to determine meaningful data will be an AI project from problem discovery, data collection, data sorting, data analysis, data interpretation, management decision-making to business intelligence to service industry change management [14].

## 2.3 Application Study for Data Visualization & Dashboard

There have been many industrial application methods and cases of digital dashboards used in various operational data monitoring and performance indicator tracking to visually present decision-making reference at home and abroad. Listed and briefly described below.

In the fiercely competitive industrial environment in the global market, the economic model has gradually become globalized and regional integration. Shipping companies are influenced by factors such as exchange rates, interest rates, fleets, shipping schedules, shipping schedules, oil prices, wars, port strikes, and turmoil. An important decision-making factor in the battle center. This research is mainly about the application of business intelligence technology, the establishment of enterprise project war situation room, the direction of collaboration and cooperation, the integration of enterprise application, including enterprise resource planning, supply key system, customer relationship management and external war situation center important decision factor information system The integration of various types of digital dashboards in the war situation room is fully presented in the form of visual charts and dashboards to help clarify and improve operational problems, improve real-time response speed and decision-making ability, and achieve pre-event prevention and post-event prevention. Reduce losses in order to provide real-time and correct decision-making and forecast information, and improve the quality of supervisors' decision-making [23].

Huge amounts of data, like crude oil, must be carefully selected and refined to produce high-value products. The most difficult problem of big data is the diversification of analysis data. If it can effectively integrate and make good use of multiple data inside and outside the enterprise, conduct interactive analysis and find the correlation between the data, it will produce great benefits. CPC Company (Taiwan, R.O.C.) decided to establish a "big data center" since Chairman Dai Qian took office in November 2016. First of all, organize the "Big Data Planning Promotion Team" to compile the data sets submitted by each unit, conduct visual analysis and design, and find that the data of each unit is all-encompassing. Secondly, in order to build a new generation of senior executives "operation and business data center", plan the company's internal big data application demonstration pilot cases - air pollution indicators, pipeline monitoring and oil tank maintenance monitoring, hoping to use big data visualization and intelligent application, Assist the supervisor to control the operation status in a timely manner, so as to make the best maintenance and operation decision [13]

Healthcare organizations have continued to invest heavily in information systems to support the operations of various departments. These systems generate large amounts of data, but fail to effectively convert this data into information that decision makers can refer to. This ineffectiveness is often attributed to strategic planning and support Lack of consistency between IT plans for operational objectives. For example, the case study of St. Joseph Mercy Oakland (SJMO) Hospital in Michigan, USA is developing the "iDashboard" that combines KPI and Dashboard to strengthen the gap between strategy and information planning, and also brings other tangible and intangible benefits; such as Enable leadership to hold employees accountable for achieving their performance goals. By displaying iDashboards in prominent locations across the hospital, such as operating room floors, dining halls, and nursing stations, SJMO Hospitals introduces transparency into the planning and monitoring process, and the need to develop KPIs and drive data collection efforts has taken root within the organization, although iDashboards support This cultural transformation plays an important role, and during the two-year study period, many KPIs have been observed to improve, but the investment is focused on technology-driven vision and actions that become the responsibility of all stakeholders[40].

In Taiwan, there is a 2,600-bed teaching hospital in the north. The COVID-19 epidemic command center was established on January 23, 2020. The person in charge presided over daily briefings and effectively formulated cross-departmental policies. The research team developed an online COVID-19 dashboard to provide medical management with visual and real-time updated data, so as to accurately and quickly grasp the epidemic/medical capacity and respond promptly and flexibly. This study describes the development architecture of the system and designs a dashboard that users can click to display graphs and data in real time. At midnight every day and every hour, the databases of the online medical information system and laboratory information system are connected to each other, and the important information related to epidemic prevention is updated in real time on the business software dashboard of the hospital network for the decision-making reference of the epidemic supervisor. It takes 8.4 hours to complete the manual calculation of all items each time, and the system only takes 16 minutes at most to click and load the data. During the peak period of the epidemic, data often needs to be updated, and manual work is simply not enough to meet the actual demand. In addition, we monitor community sustainability indicators, and efficiently carry out plan-do-check-act (PDCA) circular quality improvement measures through factor analysis to ensure the achievement of goals. In addition, patients are reminded and annotated on the clinical electronic whiteboard to help colleagues to know the distribution of patients and their isolation and protection levels [25].

The Hospital Accreditation and Healthcare Quality Council (HQC) collects data from the annual Patient Safety Culture Survey of each hospital and provides only the raw data for download for each hospital to analyze its own survey results. A dashboard was designed for each hospital department to compare their survey results. By downloading the patient safety culture survey data provided by the Medical Council and applying it to the dashboard designed by the Microsoft Excel function, each department selected the translated codes to obtain: (1) the comparison of their own department with the average and 80th percentile of the whole population for each question and each component, and the comparison of the results between groups of departments, (2) a graphical representation of the distribution of the two component coordinates between their own department and the whole population or between groups of departments, and (3) a graphical representation of the distribution of the two component coordinates between their own department and the whole population or between groups of departments. (2) a graph showing the scattering of the two two-structured coordinates between one's own department and all departments or groups of departments, (3) a graph showing the outliers among the evaluated units by clicking on the two columns of correlation coefficients and deviation indicators, and (4) a histogram showing the comparison of the scores of the evaluated units with those of other units. The raw data from the Sickness Safety Culture Survey was converted into a dashboard as an information module for quick understanding of the information.

Taiwan has been promoting the establishment of an advanced mass transportation system for nearly 20 years, including bus dynamic information, electronic tickets, digital driving recorders and other systems have been completed. The bus passenger transport industry still lacks the application of the above-mentioned large amount of data in the improvement of operation management. Based on this, this research introduces big data and information platform development technology, integrates government public information and industry operation information, and presents it in cloud technology and data visualization to analyze driving behavior, fuel consumption, credit card and other information, and develop a set of online The used "Smart Bus Operation Management Platform" provides application modules such as vehicle dynamics, driving safety, fuel saving, service, public opinion, and e-ticket analysis. Through this platform, managers of the bus passenger transport industry can instantly grasp the vehicle operation status, driving behavior and Driving safety and fuel consumption, service punctuality, passenger boarding needs, etc., have been introduced to many bus passenger transport operators in Taiwan [17].

With the advent of the post-epidemic era, it will become more and more difficult for senior managers or auditors to conduct onsite management and auditing, and the type of risk control and management by enterprises will also change. The enterprise risk control platform developed by Deloitte & Touche is through the methods of "big data analysis" and "data visualization" to analyze and diagnose and build a risk monitoring platform. Opportunities to quickly adjust and mitigate the impact, and through some leading indicators or forecasts, companies can prevent the possible damage before the risk occurs. In addition to saving the waste or loss measured in money, data analysis and visualization The results also provide various basis and opportunities for enterprises to improve and optimize, and become the best tool for enterprises to continue to maintain stable operations and create competitive advantages in the post-epidemic era. The key is to combine the manager's past effective "risk identification and assessment" experience, and refer to "enterprise process best practice" with appropriate "data analysis and visualization technology", the combination of the three can make the enterprise risk monitoring platform achieve best synergy [41].

# 3 MODEL & SOLUTION DEVELOPMENT OF PVCMS

This study is based on the research background of responding to the government's digital transformation promotion policy and the industry's digital development trend, plus years of experience in enterprise organization diagnosis and coaching, and familiarity with data processing and analysis methods as research advantages, aiming to serve small and micro enterprises with relatively few resources. Enterprise organizations are the target customers to meet the organizational productivity management needs of KPIs and project management (PjM) as the research subject, combined with MBO and performance appraisal for organizational management practices are used as research tools and methodology. Using Google's no-code smart cloud service function to integrate the patent concept of the data input/processing/analysis/visualization (D/IPAV®) workflow proposed by this researcher, Customized development of enterprise organization operation data monitoring and decision-making digital dashboard PVCMS (performance visualization cloud management system), the following are respectively explained.

## 3.1 D/IPAV®Model Building

The agile self-service business intelligence system development method constructed by Lin & Sun [26] combines the rapid data integration and interactive visual chart design functions of self-service business intelligence technology, and is based on agile development methods to help ordinary users quickly develop a self-service business intelligence system. This method has four steps: (1). namely defining business requirements, (2). obtaining and integrating relevant data, (3). designing prototype diagrams, and (4). publishing and providing feedback. These four steps are interlinked and repeated in each sprint. It also combines the method of determining the feasibility of requirements, the situation story method and database reverse engineering and other system development methods to help clarify user needs and reduce the dependence on the information department.

In addition, data visualization is the presentation of data in a graphic or chart format that makes it easier, faster, and more efficient to understand the data. Research shows that the human brain can process images 60,000 times faster than text. Therefore, with the increasing amount of data collected, business decision makers need to systematically convert a large amount of data into useful information, and data visualization will help. Presenting data analysis results at a glance, and even predicting future trends, will be more efficient than consulting a large number of spreadsheets or written reports [37].

Therefore, based on the big data analysis process and data visualization application of cloud services, this study proposes to integrate (1). data import, (2). data process, (3). data analysis and (4). data visualization four steps for D/IPAV®workflow (show as figure

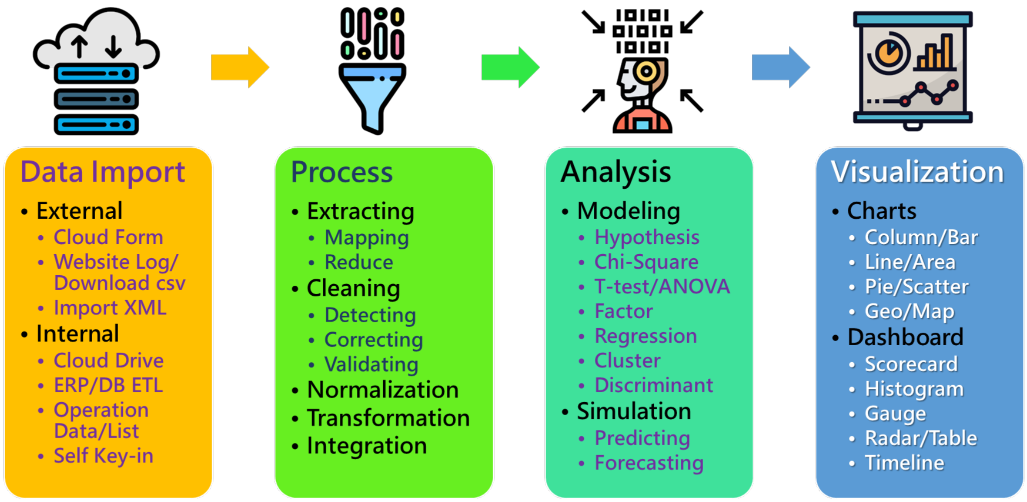


**Figure 1: D/IPAV®Workflow (Data Import/ Process / Analyze / Visualize)**

1). Using the low-cost cloud database in the internet/cloud, and can setup with automated process robots, cloud edge computing programs, etc., to input various operational data in real time or link and import it online. The cloud-based system simultaneously performs real-time calculation and analysis of operational data, and through visual design, the analysis results are effectively and instantly presented to the user/manager, so that managers at all levels and high-level decision makers can identify differences early. Then analyze the source of the problem and devise countermeasures, discuss solutions, and then solve any business problems in operation management that may occur in the operation process.

## 3.2 PVCMS Solution Development

This research team observed that No-Code/Low-Code (NC/LC) has become an emerging field of software system development in recent years, and many large information companies and new startups have invested in the establishment of various cloud platforms. After the organization is eager for a ride on digital transformation, the business opportunities for developing the platform economy are infinite; at the same time, this research team has accumulated many years of experience in corporate field coaching, and often encounters the dilemma that the management field doesn't understand information, and the information field doesn't understand management. Therefore, this research integrates the two professional fields of "information" and "management" that have already intersected. It doesn't try to solve all the management problems of enterprise organizations,but focuses on the organizations that leaders and managers care about most department and performance management issues, and plan the development path of the solution from the following three perspectives:

(1). Diagnose and build operational data indicators for departmental employee performance evaluation:

The first step in the success of performance management is to set appropriate performance indicators, strategic performance indicators that consider the overall strategic goals of the enterprise, and operational performance indicators at the department/individual level. There are differences in terms of performance management. In order to make managers and employees feel about performance management, it is necessary to connect strategic and operational performance indicators. This research team has the ability to learn industry knowledge, as well as the ability to diagnose enterprise needs. The coaching skills can be gradually established by in-depth discussions with the owners of the strategic performance indicators of the small and micro enterprises in the organization and the operational performance indicators of the operation.

(2). Accelerate the digital transformation of small and micro enterprises through the No-Code application platform:

This plan does not focus on developing a new No-Code Application Platform (NCAP), nor does it directly introduce new foreign NC/LC platforms that most domestic people do not know about, but a unique creative idea to integrate Google cloud services that are familiar and accepted by most small and micro enterprises: Google Workspace's productivity software Google Forms and Google Spreadsheets functions and chart functions for data import and analysis, and preliminary visualization presenting from linking with Google Data Studio, which is commonly used in the field of Google Analytics website flow analysis, as an advanced and integrated data visualization report, so that small and micro enterprises can have sufficient budget and full confidence to accelerate digital transformation.

(3). Integrate low-cost/stable/high-familiar Google cloud services:

This project is to use the Google account single sign-in (SSO) method to seamlessly connect various functions, plug-ins and friendly graphical interface and drag-and-drop of the Google cloud service family (Forms⟶Spreadsheets⟶Data Studio). In the small and micro enterprises with generally insufficient information manpower, it can realize the diagnosis and guidance of the establishment of KPIs of the enterprise organization, and the follow-up one-stop

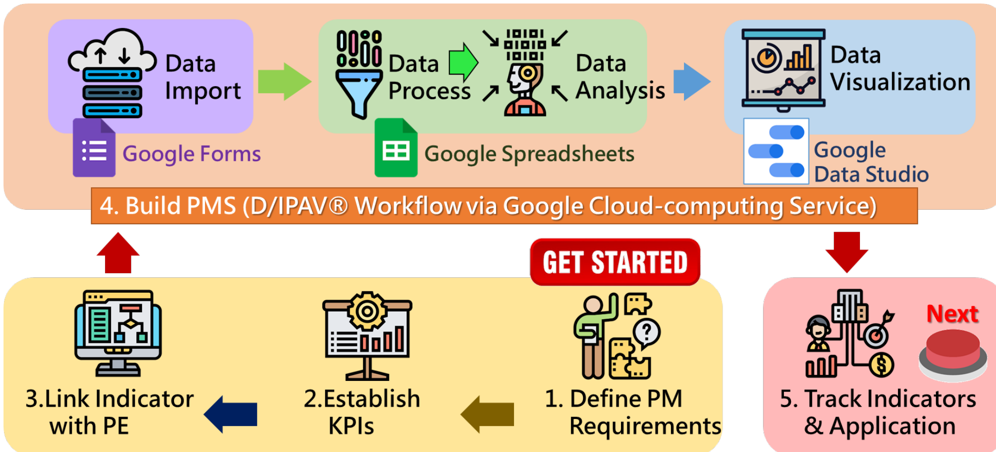


Figure 2: PVCMS Solution (Performance Visualization Cloud-computing Management System)

| | A | B | C | D | E | F | G | H | I | J | K |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Year | 111 | Month | 09 | | 111/09 | 2,500 | | | | |
| 2 | KPIs | KPIs-Department | Percentage | Performance | Objective | Ratio | Bonus | Weighted Ratio | Department Contribution | Bonus-Department | Minus |
| 3 | A.Factory 50% | A1.Productivity | 20% | 10,974,050 | 10,930,000 | 100.4% | 502 | 46.7% | 1,167 | 2,473 | |
| 4 | | A2.Productivity/per Emp. | 30% | 322,766 | 364,000 | 88.7% | 665 | | | | |
| 5 | B.Distribution 6% | B1.Accurace/Deliver | 3% | 100.0% | 100.0% | 100.0% | 75 | 5.9% | 148 | 2,609 | |
| 6 | | B2.Accurace/Inventory | 3% | 97.0% | 100.0% | 97.0% | 73 | | | | |
| 7 | C.Store-1 24% | C1.Revenue/Resturant | 8% | 1,239,010 | 1,300,000 | 95.3% | 191 | 24.1% | 603 | 2,664 | |
| 8 | | C2.Revenue/Retail | 8% | 2,156,098 | 2,000,000 | 107.8% | 216 | | | | |
| 9 | | C3.Revenue/per Emp. | 6% | 323,961 | 330,000 | 98.2% | 147 | | | | |
| 10 | | C4.Negative review/Google | 2% | 0.0 | 0.0 | 100.0% | 50 | | | | |
| 11 | D.Store-2 20% | D1.Revenue/Resturant | 5% | 935,099 | 1,100,000 | 85.0% | 106 | 19.6% | 490 | 2,597 | |
| 12 | | D2.Revenue/Retail | 7% | 1,450,697 | 1,300,000 | 111.6% | 195 | | | | |
| 13 | | D3.Revenue/per Emp. | 6% | 315,999 | 342,000 | 92.4% | 139 | | | | |
| 14 | | D4.Negative review/Google | 2% | 0.0 | 0.0 | 100.0% | 50 | | | | |
| 15 | | 合計 | 100% | | | | | 96.3% | | | |
| 16 | E.Admin+10% | Average All | 10% | | | 9.6% | 241 | 9.6% | 2,649 | | |
| 17 | | SUM | 110% | | | 9.6% | 2,649 | 106.0% | | | |

Figure 3: A sample image of case dashboard for PVCMS Solution

service that has the ability to collect data, automatic statistics and visual presentation for a long time. The handover of Google accounts and files sharing are used to transfer the system painlessly and facilitate multi-party synchronization during the tutoring period. Considering the budget affordability of small and micro enterprises for importing digital platforms, they use low-cost/stable/high-cost The familiar Google cloud service is integrated into a cloud performance management system with real-time computing linkage and visual presentation.

Thus, this research takes the PVCMS (Performance Visualization Cloud-computing Management System) as the core product, and develops a set of indicators establishment, analysis and visual monitoring that meet the performance management needs of the industry. The diagnosis and coaching model is the solution, as the standard operating process followed by subsequent actual commercialization, and as a quick introduction template for various performance management visualization modules. Its operation process is mainly divided into five stages: (1). Define performance management requirements, (2). Establish KPIs, (3). Link Indicator with performance evaluation, (4). Build performance management system, (5). Track Indicators and decision-making application. Drawn as shown in Figure 2 below.

## 4 DISCUSSIONS

This research is aimed at the practical performance management needs of small and micro enterprises with relatively limited resources. For example, senior management cadres generally lack knowledge of digital transformation, lack of professional information talents, and low information budgets, which make it difficult to synchronize digital performance when industrial technology changes. Provide a sample case dashboard as figure 3.

Management tools are imported and remain manual jobs that are prone to errors, wastes manpower and fails to reflect performance in a timely manner. Based on this, this research integrates the Google Workspace and Data Studio no-code platforms code under Google, which is a leading cloud service provider, supplemented by a professional consultant team to provide diagnosis and discussion of performance management indicators, and follow-up performance visualization cloud management system. Establish and long-term tracking performance evaluation management system for organizations / departments and employees to provide a set of sustainable performance management solutions. In addition, the expected difficulties and solutions are discussed as follows:

(1) Performance management involves sensitive information of the enterprise organization, and the initial practice case needs to gain enough trust from the owner

This study is to develop a cloud-based solution for the diagnosis and coaching model of indicators establish, analysis and visualization monitoring of performance management needs in the industry, especially in the selection and proportional allocation of key performance indicators, which need to be discussed repeatedly, and even after the system is launched, It is still necessary to continuously monitor performance changes, and it is bound to have long-term exposure to the performance-sensitive information of the enterprise organization. Therefore, it is very necessary to keep confidential the sensitive information of the relevant human resources, finance and various sales and production management data of the enterprise organization, through the formation of the research team's confidentiality habits and the signing of written confidentiality clauses to build trust with the case enterprise organization.

(2). The No-Code platform has different practices, and it is difficult to convert and migrate once it is fixed: This research clearly understands that the reason why the No-Code application platform can achieve App functions without writing code is the product characteristics of SaaS (Software as a Service). Build the bottom layer of the system with rich operation-related functions. Therefore, if the SaaS service is no longer used in the future, the previously set and designed application functions usually can't be taken away, or can't be easily imported to another service platform. So, the sunk cost of software development is very high, which is also the ability of the No-Code application platform, which mainly relies on a subscription-based charging mechanism. Therefore, this research integrates Google cloud services, which are familiar and accepted by most small and micro enterprises, and uses Google Spreadsheets of Google Workspace as the core database. An open and flexible spreadsheet format will be the best solution for subsequent borrowing from different platforms.

## 5 CONTRIBUTIONS & RECOMMENDATIONS

This research proposes that the No-Code platform has become a powerful tool for small and micro enterprises that lack information and human resources on the road to digital transformation. If it can be effectively used in the field of performance management, the following three expected impacts can be achieved.

(1). The No-Code platform can accelerate the digital transformation of small and micro enterprises

(2). Lowering the technical threshold allows capital to develop the management system by itself

(3). Transparent performance indicator monitoring to improve the effectiveness of performance management.

Therefore, the main contributions of this research lie in: importing Google's no-code cloud service functions with data-oriented empirical management methods, integrating D/IPAV®workflows, and customizing the development of enterprise organization operational data monitoring and decision-making digital dashboard with PVCMS solution. It is expected to assist small and micro enterprises with relatively limited talent and financial resources to provide organizational/departmental and employee performance and productivity management coaching services, help clarify organizational productivity management issues, and then use a visualization cloud management system linked to KPIs to improve organizational performance management as a way to improve the operational performance of small and micro enterprises in the face of digital transformation, it can also transfer technology to management consulting companies after the completion of system commercialization to strengthen their industrial competitiveness.

Especially in the post-epidemic era, remote work and provision of services have become popular. Rapidly developing required application systems and apps through the No-Code platform will be an important digital skill in the path of digital transformation. Starting with the performance evaluation management system that every business owner attaches great importance to, it will be able to set an example and lead the trend of the overall development of the national public and private sectors in digitalization, digital optimization and digital transformation.

At the same time, this research is innovative, and more empirical research is still needed to verify the academic value and practical feasibility of the research results. Therefore, this research attempts to make suggestions for follow-up research:

(1). Use the agile "Kanban" development method for No-Code system development:

It is recommended that subsequent researchers can develop in a No-Code way that is familiar with the business operation process. The process and practice are relatively simple, and follow the spirit and principles of agile development. In agile development, the Kanban development method is adopted. Kanban uses etymology From Japanese, it comes from Toyota Lean Production, which is famous for its JIT (Just in Time) timely production. Agile development borrows the visual management concept behind it, and uses the Kanban form as the visualization of the project management process. As the core of the daily discussion of the agile team, it also simultaneously understands the work status of other members and the progress of their own related work, and becomes one of the methods to solve the communication barriers between teams. There will be user stories that may need to be done, and the development process can be managed and the details and history of the process can be recorded. Visualize the status and number of to-do items, and identify bottlenecks to improve.

(2). Follow-up can use organizational case study method to verify the solution:

It is recommended that follow-up researchers take small and micro enterprise organizations with limited resources as an empirical organization case, use the Google no-code smart cloud service function, integrate the D/IPAV®Workflow proposed in this study, and customize the development of enterprise organization operation data monitoring and decision-making data. PVCMS Solution on the dashboard, and detailed observation and recording of the reactions and experiences of organizational cases before, during and after the development of the PVCMS solution, supplemented by individual interviews with organizational supervisors and persons in charge and qualitative research of the project meeting, comprehensively put forward the overall case-by-case results.

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The Taiwan (R.O.C.) invention patent for D/IPAV®Workflow in this study has been filed and is currently under review.

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